Eigenvalues

a short primer

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1 Definitions

Let A be an n × n matrix over the real or complex numbers. A non-zero vector v is an eigenvector of A when multiplying by A only rescales it. The scale factor is the corresponding eigenvalue.

$$A\,v = \lambda\,v, \quad v \neq 0 \tag{1.1}$$

Geometrically, an eigenvector spans a line that A maps onto itself. The eigenvalue λ records how that line is stretched, flipped, or collapsed.

1.1 The characteristic polynomial

Rewriting (1.1) as (A − λ I) v = 0 shows that a non-zero solution exists exactly when A − λ I is singular, that is, when its determinant vanishes.

$$p(\lambda) = \det(A - \lambda\, I) = 0 \tag{1.2}$$

The polynomial p has degree n, so A has exactly n eigenvalues counted with multiplicity over the complex numbers.

> **Definition — spectrum**
> The set of eigenvalues of A is called its spectrum and is written σ(A). The largest absolute value in σ(A) is the spectral radius.

1.2 A worked example

Take the 2 × 2 matrix with rows (2, 1) and (1, 2). Its characteristic polynomial is

$$(2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) \tag{1.3}$$

so σ(A) = {1, 3}. The eigenvector for λ = 3 is (1, 1): the matrix stretches the diagonal by three. The eigenvector for λ = 1 is (1, −1): the anti-diagonal is left unchanged.

2 Properties

2.1 Trace and determinant

Comparing coefficients of p(λ) with the product of its linear factors gives two identities that are often the quickest sanity check on a calculation.

$$\operatorname{tr} A = \Sigma\ \lambda_i\ , \qquad \det A = \Pi\ \lambda_i \tag{2.1}$$

In the example above the trace is 4 = 1 + 3 and the determinant is 3 = 1 · 3.

2.2 Symmetric matrices

When A equals its transpose every eigenvalue is real, and eigenvectors belonging to distinct eigenvalues are orthogonal. This is the spectral theorem, and it is why symmetric matrices can always be diagonalised by a rotation.

> **Theorem — spectral theorem**
>
> If A is real and symmetric there is an orthogonal matrix Q and a real diagonal matrix D with A = Q D Q^{T}. The columns of Q are orthonormal eigenvectors of A.

2.3 Powers and stability

If A = Q D Q^{-1} then A^{k} = Q D^{k} Q^{-1}, so repeated application of A is governed by the powers of its eigenvalues. The iteration x_{k+1} = A x_{k} tends to zero for every start exactly when the spectral radius is below one.

$$\rho(A) < 1 \quad \Rightarrow \quad A^{k}\, x \to 0 \quad \text{as} \quad k \to \infty \tag{2.2}$$

3 Computing eigenvalues

3.1 Power iteration

Starting from a random vector and repeatedly multiplying by A, then normalising, converges to the eigenvector of the eigenvalue with largest absolute value, provided that eigenvalue is strictly dominant. The convergence rate is the ratio of the two largest eigenvalues in absolute value.

$$x_{k+1} = A\, x_k \,/\, |A\, x_k| \tag{3.1}$$

3.2 The QR algorithm

Practical libraries use the QR algorithm: factor A = Q R, form R Q, and repeat. Each step is a

similarity transform, so the spectrum is preserved, and the iterates converge to an upper-triangular matrix whose diagonal holds the eigenvalues.

Exercise

Find the eigenvalues of the matrix with rows (0, 1) and (–1, 0). Why are they not real, and what does the matrix do geometrically?

Further reading

G. Strang, Introduction to Linear Algebra, chapter 6. L. N. Trefethen and D. Bau, Numerical Linear Algebra, lectures 24 to 29.